

Week-6.

$$f: \mathbb{R} \rightarrow \mathbb{R}.$$

$$f(x) = \{x\} \leftarrow \text{fractional part of "x"}$$

$$\text{Dom}(f) = \mathbb{R}, \quad \text{Range}(f) = \underline{[0, 1)}$$

$$\{x\} := x - \lfloor x \rfloor.$$

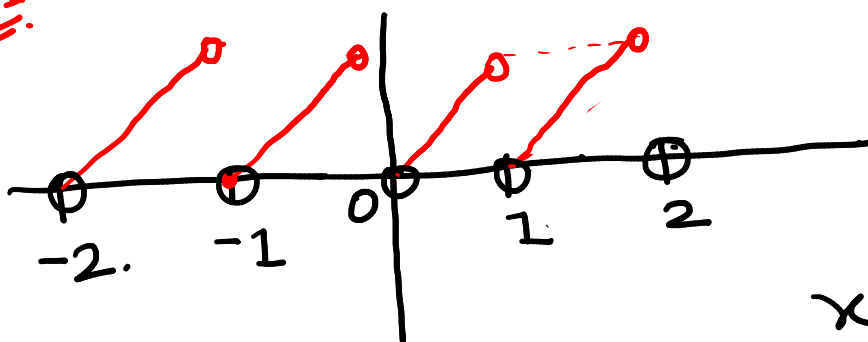
$$f(1) = \{1\} = 1 - \lfloor 1 \rfloor = 1 - 1 = 0$$

$$f(1.7) = \{1.7\} = 1.7 - \lfloor 1.7 \rfloor = 0.7$$

$$f(-2) = \underline{0}$$

$$\begin{aligned} f(-2.1) &= \{-2.1\} = -2.1 - \underline{\underline{\lfloor -2.1 \rfloor}} \\ &= -2.1 - (-3) \\ &= \underline{\underline{0.9}} \end{aligned}$$

$$\begin{aligned} f(0.9) &= \underline{\underline{0.9}} \\ f(-0.9) &= \underline{\underline{0.1}} \end{aligned}$$



$$x - \lfloor x \rfloor \geq 0$$

$$\bullet \quad f(x) = \lfloor x \rfloor$$



greatest integer which is less than or
equal to "x"

$$f(1) = 1$$

$$f(1.9) = 1$$

$$\underline{\underline{-1}}, \underline{\underline{0}}, \underline{\underline{1}}, 2, 3, -$$

$$f(1.3) = 1$$

$$f(2) = 2$$

$$\underline{\underline{1}}, 1.3, 2$$

$$f(-1) = -1,$$

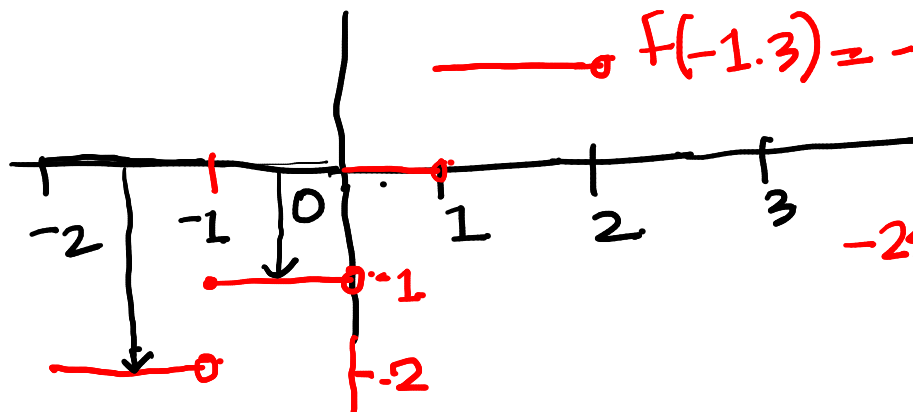
$$\underline{f(-1.2) = -2.}$$

$$\underline{-2}, \underline{-1.2}, -1, 0$$

$$f(-1.3) = -2.$$

$$f(-2) = -2$$

$$-2 \leq x < -1$$



$$\log_a 1 = 0 \Leftrightarrow a^0 = 1$$

$$\text{If } a \neq 0$$

$$\log_a x$$

$$\begin{matrix} a \neq 1 \\ a > 0 \end{matrix}$$

$$\log_a x - \log_x 1$$

$$\begin{matrix} \uparrow \\ \text{Inverse of} \\ f(x) = a^x \end{matrix}$$

$$\text{If } a = 1 \quad f(x) = 1^x \text{ - Constant function}$$

Imparti

$$\log_a x = y \Leftrightarrow a^y = x$$

$$x \xrightarrow{\text{exp.}} (a^x) = m$$

$$x \xleftarrow{\log_a} m$$

$$\log_a m = x$$

$$\underline{m = a^x}$$

Q2

$$f(x) = \log_2(2-x) \Leftrightarrow \boxed{2-x > 0} \Rightarrow 2 > x.$$

$$\Leftrightarrow \boxed{x < 2}$$

$$\log_2 x \leftarrow (0, \infty)$$

$$\text{iff } x > 0$$

$$2 - x + x > x \Rightarrow 2 > x.$$

$$\log_{10} x$$

Q3

$$\log_{10} \left[(\log_{10} x)^2 - 8 \log_{10} x + 15 \right]$$

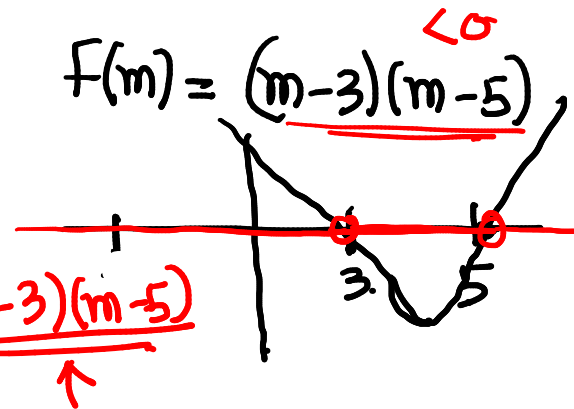
$$(\log_{10} x)^2 - 8 \log_{10} x + 15 \geq 0$$

$$\boxed{m = \log_{10} x}$$

$$\Rightarrow m^2 - 8m + 15 > 0$$

$$\Rightarrow \boxed{(m-3)(m-5) > 0}$$

$$m \in (-\infty, 3) \cup (5, \infty). \quad f(m) = \underbrace{(m-3)(m-5)}$$



$$\Rightarrow \log_{10} x \in (-\infty, 3) \cup (5, \infty).$$

$$\log_{10} x \in (5, \infty) \Leftrightarrow 5 < \log_{10} x$$

$$\Leftrightarrow x > 10^5$$

$$\Leftrightarrow x \in (10^5, \infty)$$

$$\log_{10} x \in (-\infty, 3) \Leftrightarrow \log_{10} x < 3.$$

If $x < 0$

$$\Leftrightarrow 0 < x < 10^3$$

$$(0, 10^3) \cup (10^5, \infty)$$

$$f(x) = \log_{\frac{x-1}{x+1}} \left(\frac{x^2+1}{x^2-1} \right)$$

$\frac{x-1}{x+1} = a$

~~$a \neq 1$~~ $\wedge$ $a > 0$

$$\frac{x-1}{x+1} > 0$$

$$x-1 < x+1$$

$$\Rightarrow \frac{x-1}{x+1} < 1$$

$$(-\infty, -1) \cup (1, \infty)$$

$$a = \frac{x-1}{x+1} = \overset{m}{x-1} \times \overset{n}{\frac{1}{x+1}} > 0$$

$$m \times n > 0$$

$$\Rightarrow x > 1 \Rightarrow (1, \infty)$$

$$0 < a = m \times n$$

$\bullet$ $m > 0, n > 0 \Rightarrow x-1 > 0$ $\frac{1}{x+1} > 0$

$\bullet$ $m < 0, n < 0 \Rightarrow \frac{x-1}{x+1} < 0$ $\frac{1}{x+1} < 0$

$x < 1$, $\frac{1}{x+1} < 0 \Leftrightarrow x+1 < 0$

$(-\infty, 1)$ $\Leftrightarrow x < -1$ $(-\infty, -1)$

$$\Rightarrow (-\infty, -1)$$

$x \in (-\infty, -1) \cup (1, \infty) = a > 0$

$$\frac{x^2+1}{x^2-1} > 0$$

$$x^2+1 \geq 1$$

$$[-1, 1]$$

$$x^2-1 > 0 \Leftrightarrow |x| > 1$$

$$\underline{x < -1 \text{ } \& \text{ } x > 1}$$

$$f(x) = \log(- (x e^{3x} - 27x))$$

$$- (x e^{3x} - 27x) > 0$$

$$\Leftrightarrow x e^{3x} - 27x < 0$$

$$x \cdot (e^{3x} - 27) < 0$$

$$\cancel{x < 0 \& \& e^{3x} - 27 > 0}$$

$$x < 0$$

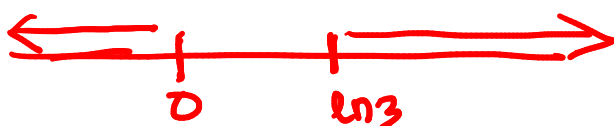
$$e^{3x} - 27 > 0$$

$$\Rightarrow e^{3x} > 27$$

$$\Rightarrow x > \ln 3$$

$$\Rightarrow x \in (\ln 3, \infty)$$

$$(-\infty, 0)$$



$$x > 0 \& \& e^{3x} - 27 < 0$$

$$(0, \infty)$$

$$e^{3x} < 27$$

$$\Rightarrow \ln_e e^{3x} < \ln_e 27$$

$$\Rightarrow 3x < \ln_e 3^3$$

$$\Rightarrow 3x < 3 \ln_e 3$$

$$\Rightarrow x < \ln_e 3$$

$$(0, \ln 3)$$

$$15^{\overline{72}} = 15^{49}$$

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$$1 \leq \quad < 10$$

9 ← two digit

$$\log_{10}$$

$$\begin{array}{l} \underline{10} \leq \quad < \underline{100} \\ \Rightarrow \log_{10} \leq \quad < 2 \\ \Rightarrow \underline{1} \leq \quad < 2 \end{array}$$

$$1 \leq \log_{10} a < 2$$

$$\lfloor \log_{10} a \rfloor + 1 = \underline{2}$$

$$100 \leq a < 1000$$

$$\Rightarrow 2 \leq \log_{10} a < 3$$

$$\lfloor \log_{10} a \rfloor + 1 = \underline{3}$$

$$\lfloor \log_{10} 15^{49} \rfloor + 1$$

$$* = 49 \times \log_{10} 15 \quad \lfloor * \rfloor + 1$$

$$\ln(e^x - e^{-x}) <$$

$$e^x - e^{-x} > 0$$

$$\Rightarrow e^x - 1/e^x > 0$$

$$e^x > 0$$

$$\begin{array}{l} e^{2x} - 1 > 0 \Leftrightarrow \frac{e^{2x} - 1}{e^x} > 0 \\ \Downarrow \\ e^{2x} > 1 \Rightarrow \ln e^{2x} > \ln 1 \end{array}$$

$$\Rightarrow 2x > 0$$

$$\underline{\underline{x \in (0, \infty)}}$$

$$\Rightarrow \underline{\underline{x > 0}}$$

★ $\ln(e^x - e^{-x})$ is Incre.

$\hookrightarrow F(x) = \underline{e^x - e^{-x}}$ - Increasing $\Leftrightarrow F(x) - F(y) > 0$

$$\text{If } x > y \Rightarrow \boxed{F(x) > F(y)}$$

$$\Rightarrow \ln(F(x)) > \ln(F(y))$$

$$\underline{\underline{F(x)}} \quad \underline{\underline{F(y)}}$$

$$\log_2(\log_2(\log_2 x)) > 0 \quad \leftarrow \log_2(x)$$

$$2^{\log_2 x} = x.$$

f - one-one
 g - one-one
 $f \circ g$ - is also one

$$\begin{aligned} \log_2 \log_2 x &= 1 \\ \Rightarrow \log_2 x &= 2 \\ \Rightarrow \underline{\underline{x}} &= \underline{\underline{4}} \end{aligned}$$

$$\log_2(\log_2 x) > 0$$

$$\Rightarrow 2^* > 2^0 = 1$$

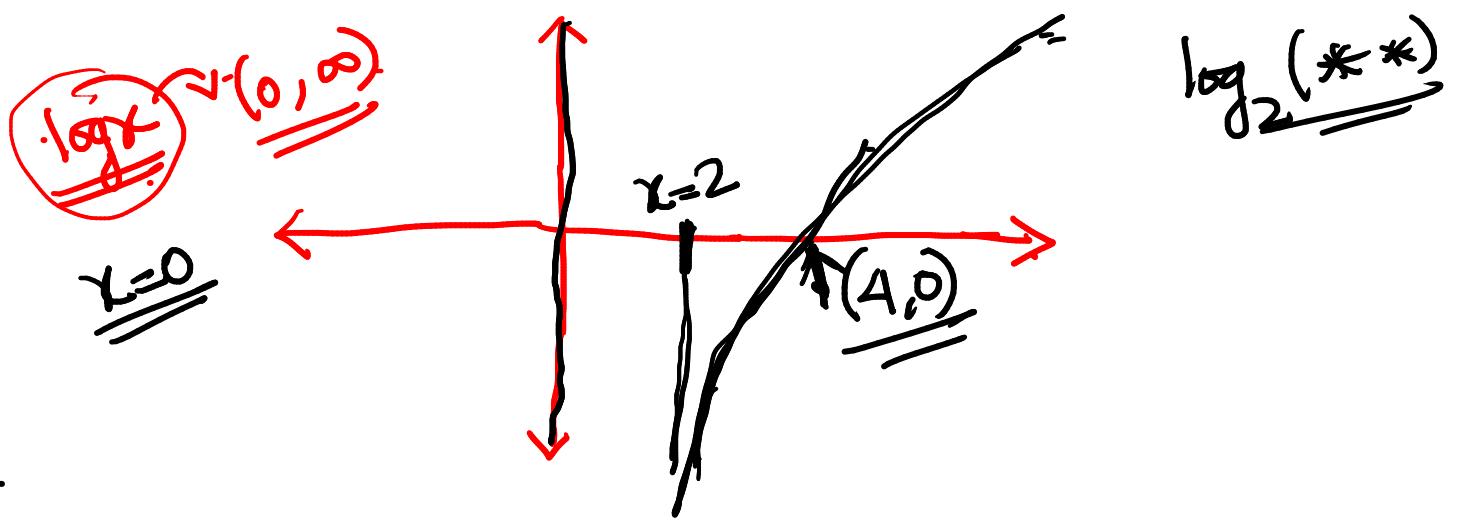
$$\Rightarrow \log_2 x > 1$$

$$\Rightarrow \underline{\underline{x > 2}}$$

$$f(x) = \log_2 x \quad \text{one-one}$$

$$g(x) = \log_2 x \quad \text{one}$$

$$(f \circ g)(x) = \underline{\underline{\log_2(\log_2 x)}} \quad \text{one-one}$$



- ✓ If $x \in \text{Dom}(g)$ then $x \in \text{Dom}(f \circ g)$
 - ✓ $\{x \mid g(x) \in \text{Dom}(f)\}$
 - ✓ $x \in \text{Dom}(f)$
- $(f \circ g) = \underline{\underline{(-2, \infty)}}$

• $f(x) = 1/x^2$

$g(x) = \frac{\sqrt{x+2}}{5x}$ $g(0)$

• $(f \circ g) = I$ & $(g \circ f) = I$

$(f \circ g)(x) = f(1 - 1/x) = \frac{1}{1 - 1/x - 1} = \underline{\underline{-x}}$

• $\ln a - \ln b = \ln\left(\frac{a}{b}\right)$

$\ln\left(\frac{x^2}{4}\right) - \ln(x-1) = \log_5 5 = 1$

$\Rightarrow \ln\left(\frac{x^2}{4(x-1)}\right) = \ln_e e$

$$\Rightarrow \frac{x^2}{4(x-1)} = e. \Rightarrow \boxed{x^2 = 4e(x-1)}$$

+ve Real

$$f(x) = \log_{10}(x - \lfloor x \rfloor) \rightarrow \text{has an inverse}$$

$$f(1.2) = \log_{10}(\underline{1.2 - \lfloor 1.2 \rfloor}) > 0. \quad f(1.2) = \log_{10} 0.2$$

$$D = \mathbb{R} \setminus \underline{\underline{\mathbb{Z}}} = \log_{10} 0.2 \quad \bullet \quad D \text{ is } \quad f(2.2) = \log_{10} 0.2$$

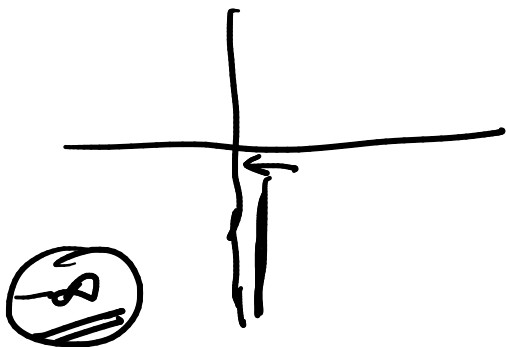
$$\underline{\underline{\mathbb{R} \setminus D}}$$

$$\mathbb{R} \setminus D = \mathbb{Z} \begin{matrix} 1.2 \\ 2.2 \end{matrix} \rightarrow \begin{matrix} \nearrow \\ \nwarrow \end{matrix}$$

$$\underline{\underline{A \setminus B}} = \{x \in A \mid x \notin B\}$$

$$\underline{\underline{\mathbb{R} \setminus D}} = \{x \in \textcircled{\mathbb{R}} \mid x \notin \textcircled{D}\} = \textcircled{\mathbb{Z}}$$

$$\log(\underline{\underline{x - \lfloor x \rfloor}})$$



$$0 < x - \lfloor x \rfloor < 1$$

$$\underline{\underline{-\infty < \log_{10}(x - \lfloor x \rfloor) < \log_{10} 1}}$$

$$\log_{10} 1$$

$$1, 2, 3, \dots$$

$$\textcircled{0}$$

~~$$f(x) = \log_{10}(\underline{x - LxL})$$~~

$$f(0.9) = \log_{10} 0.9$$

$$\log_{10} 1 = 0$$

